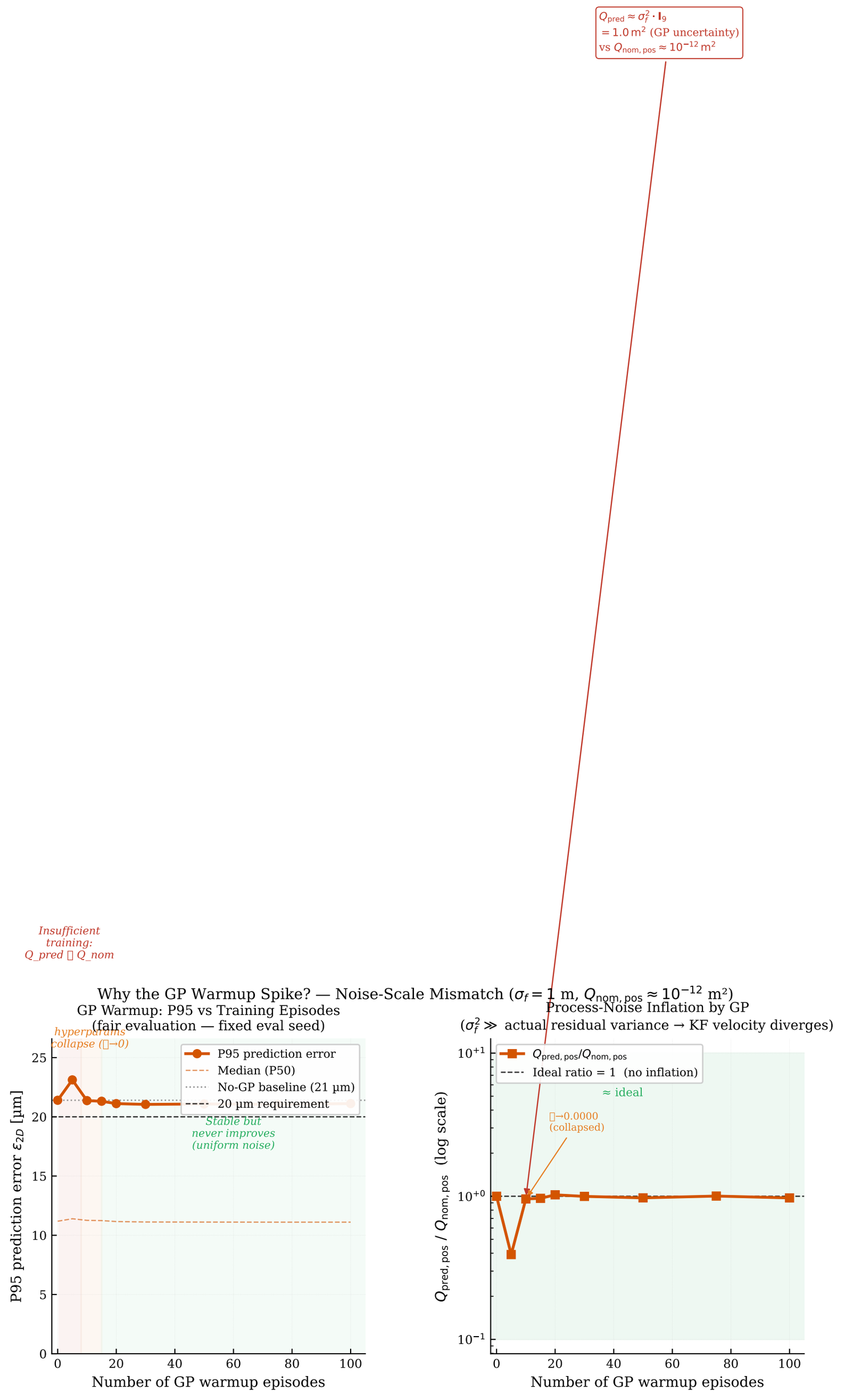
$Q_{pred} \approx \sigma_f^2 \cdot I_9$
$= 1.0\,m^2$ (GP uncertainty)
vs $Q_{nom,pos} \approx 10^{-12}\,m^2$
Insufficient training: Q_pred ≪ Q_nom
Why the GP Warmup Spike? — Noise-Scale Mismatch ($\sigma_f = 1$ m, $Q_{nom,pos} \approx 10^{-12}$ m²)
GP Warmup: P95 vs Training Episodes
(fair evaluation — fixed eval seed)
hyperparams collapse (ℓ→0)
P95 prediction error
Median (P50)
No-GP baseline (21 µm)
20 µm requirement
Stable but never improves (uniform noise)
P95 prediction error ε_{2D} [µm]
Number of GP warmup episodes
Process-Noise Inflation by GP
($\sigma_f^2 \gg$ actual residual variance → KF velocity diverges)
$Q_{pred,pos}/Q_{nom,pos}$
Ideal ratio = 1 (no inflation)
≈ ideal
ℓ→0.0000 (collapsed)
$Q_{pred,pos}$ / $Q_{nom,pos}$ (log scale)
Number of GP warmup episodes